ShutongChen

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Ph.D. in Computer Science and Technology

Ph.D. candidate in Computer Science and Technology at the University of Technology Sydney, expected to graduate in February 2027. Research and engineering experience focus on LLM fine-tuning, LoRA/PEFT, model merging, RAG/Memory, Tool-use/Function Calling, and model evaluation. Familiar with PyTorch, Hugging Face Transformers, PEFT, and LLM experiment pipelines, with hands-on experience in fine-tuning LLaMA /Qwen/DeepSeek models, multi-LoRA fusion, memory-augmented generation, and instruction-following evaluation. First-author publications at ICML and AAAI, with additional NeurIPS-related work under submission. Seeking LLM Engineer / Algorithm Engineer roles.

Technical Skills

- **Programming & Frameworks**: Python, PyTorch, Hugging Face Transformers, PEFT, vLLM
- **LLM Training**: LoRA/QLoRA, SFT, Prompt Tuning, Model Merging, GRPO basics
- **LLM Applications**: RAG, Memory System, Tool-use, Function Calling, Agent Workflow
- **Evaluation**: ROUGE, Exact Match, Recall@K, ablation study, failure analysis

Education

- **University of Technology Sydney** Faculty of Engineering and Information Technology Ph.D. May 2023 – Feb. 2027 (Expected)
  - Fully funded international Ph.D. student
- **Xidian University (China)** School of Artificial Intelligence M.Eng. Sep. 2019 – Jun. 2022
  - Intelligent Science and Technology.
- **Xidian University (China)** School of Artificial Intelligence B.Eng. Sep. 2015 – Jun. 2019
  - Intelligent Science and Technology; ranked in the top 15% of the program.

Publications

- **Personalized Additive Modeling for Multi-level Federated Learning**. ICML 2026, first author.
- **FedMerge: Federated Model Merging for Personalization**. AAAI 2026, first author.
- **Federated Parametric Memory Learning: Retrieving Shareable Tokens for Personalized Generation**. NeurIPS 2026 under submission, first author.
- **Spatial Pooling Graph Convolutional Network for Hyperspectral Image Classification**. IEEE TGRS, second author.

Selected Projects

- **Multi-task LLM LoRA Fine-tuning and Dynamic Adapter Composition** 2023.05 – 2024.05
  - **Related Work**: **Personalized Additive Modeling for Multi-level Federated Learning**. ICML 2026, first author.
  - **Project**: Built a multi-level adapter composition method for multi-task LLM fine-tuning, addressing the limitation that a single LoRA adapter may not cover diverse instruction-following tasks.
  - **Implementation**: Developed an end-to-end pipeline based on Qwen3-4B and LoRA, covering level-wise training, adapter freezing, validation-based pruning, checkpoint selection, loss/ROUGE logging, and batch evaluation.
  - **Method**: Used a boosting-like residual composition strategy, first learning general modules and then adding task-related modules only when validation loss improves.
  - **Results**: Achieved loss 1.040 and ROUGE-1 59.25% on 71 FLAN-T5-derived tasks, outperforming single-LoRA average training, local LoRA fine-tuning, and MoE-style multi-LoRA baselines.
- **LLM Model Merging and Multi-LoRA Fusion Framework** 2024.06 – 2025.01
  - **Related Work**: **FedMerge: Federated Model Merging for Personalization**. AAAI 2026, first author.
  - **Project**: Studied how to merge multiple LoRA adapters / expert models into a single deployable model

for multi-task instruction-following scenarios, reducing the extra inference cost of loading multiple experts.
  - **Implementation**: Implemented LoRA module management, weight merging, merging-weight optimization, checkpoint selection, and automated batch evaluation using PyTorch and Hugging Face PEFT.
  - **Method**: Treated LoRA adapters as reusable capability modules and generated a merged model through learnable merging weights updated from merged-model training signals.
  - **Results**: On 8 FLAN task groups with LLaMA3-8B + LoRA, achieved ROUGE-1 74.86%, improving over the MoE-style multi-LoRA baseline of 73.77% by 1.09 points; each task only needs one merged LoRA module at inference time.
- **RAG-like Parametric Memory for Tool-use / Function Calling** 2025.10 – 2026.02
  - **Related Work**: **Federated Parametric Memory Learning: Retrieving Shareable Tokens for Personalized Generation**. NeurIPS 2026 under submission, first author.
  - **Project**: Explored compressing external knowledge and tool-related capabilities into lightweight memory tokens, reducing the reliance of text-based RAG on long-context memory.
  - **Implementation**: Built a memory-augmented generation pipeline based on Llama 3.2 3B Instruct, freezing the LLM backbone and training only additional token embeddings.
  - **Method**: Represented task/tool information as retrievable memory tokens; the model first retrieves memory tokens and then generates answers or function-call formatted outputs.
  - **Results**: In ToolGen/ToolBench-style function calling settings, FedToM achieved 0.875/0.870 EM using around 3 memory tokens; with LoRA, EM improved to 0.920/0.930 while reducing text memory length.
- **LLM Post-training, Inference Deployment, and Agent-related Practice** 2025.02 – 2025.09
  - Ran GRPO/post-training experiments on GSM8K using DeepSeek-R1-Distill-Qwen-1.5B, recording reward curves, generation length, and training logs.
  - Deployed vLLM inference service and completed local/server-side model loading, OpenAI-compatible API calls, and batch request testing.
  - Implemented a basic memory system with short-term memory, long-term memory, hierarchical memory, and memory retrieval in agent-like environments.
  - Studied function calling, Multi-Agent workflow, KV cache / prefix cache reuse, and model editing methods such as ROME/MEMIT.
- **Graph Convolutional Network for Remote Sensing Classification** 2019.09 – 2021.11
  - **Related Work**: **Spatial Pooling Graph Convolutional Network for Hyperspectral Image Classification**. IEEE TGRS, second author; independently led the main experimental process.
  - Built pixel patch graphs for hyperspectral images and used spatial pooling graph convolution to aggregate local contextual information.
  - Outperformed mainstream CNN baselines on multiple hyperspectral image classification benchmarks.

Reviewing & Honors

- Served as a reviewer for ICML, NeurIPS, ICLR, AISTATS, AAAI, TNNLS, and other conferences/journals.
- Academic Excellence Scholarships at Xidian University (China): Third Prize in 2017, Second Prize in 2018, Second Prize in 2021.